

+90/100

4.3 PROBLEMS 1-11

4. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 20 \end{bmatrix}$

$$T = [C^2 + (C+D-8)^2 + (C+3D-8)^2 + (C+4D-20)^2]$$

$$\frac{\partial T}{\partial C} = 2C + 2(C+D-8) + 2(C+3D-8) + 2(C+4D-20) = 0$$

$$\frac{\partial T}{\partial D} = 2(C+D-8) + 6(C+3D-8) + 8(C+4D-20) = 0$$

$$\Rightarrow C + C + D - 8 + C + 3D - 8 + C + 4D - 20 = 0$$

$$4C + 8D = 36$$

$$\Rightarrow C + D - 8 + 6C + 18D - 48 + 8C + 32D - 160 = 0$$

$$13C + 50D = 112$$

$$\Rightarrow \begin{bmatrix} 4 & 8 \\ 13 & 50 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

4. $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 3 & 0 \\ 1 & 4 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 20 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 4 & 8 & -26 \\ 8 & 26 & -42 \\ 26 & 42 & -38 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 100 \end{bmatrix}$$

In figure 4.0b, there is a new vector $a_3 = (0, 1, 1, 6)$ that produces $0 = C a_1 + D a_2 + E a_3$, possibly widening the gap between b and p - the error.

Iffy... but ok. $b = \begin{bmatrix} 0 \\ 8 \\ 20 \end{bmatrix}$

2/6. $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 3 & 0 \\ 1 & 4 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} C \\ D \\ E \end{bmatrix}$

$$A^T A = \begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 42 \\ 26 & 42 & 38 \end{bmatrix} \rightarrow A^T b = \begin{bmatrix} 36 \\ 112 \\ 100 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 42 \\ 26 & 42 & 38 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 100 \end{bmatrix}$$

$$C = 2; D = -\frac{3}{2}; E = -\frac{3}{2}$$

$$2 - \frac{3}{2}x - \frac{3}{2}y = 2$$

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \rightarrow A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

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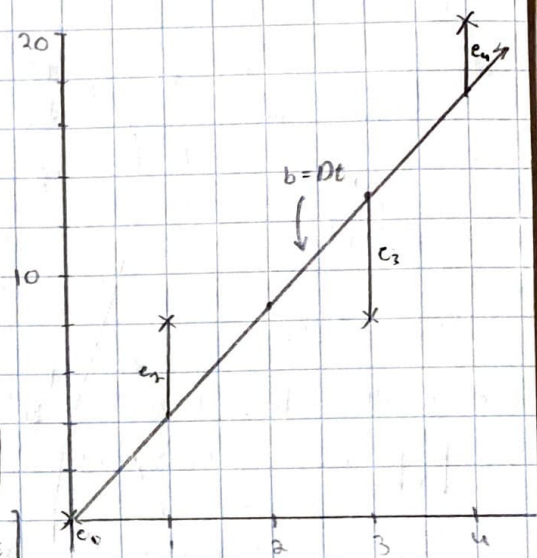
$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \rightarrow A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$C = 2; D = -\frac{3}{2}; E = -\frac{3}{2}$$

$$2 - \frac{3}{2}x - \frac{3}{2}y = 2$$



Right Answer using Method (left misspelled)

Since this plane is the only of form $Cx + Dy + Ez$ that crosses the origin, it must be the plane $x = 0, y = 0$ when plugging $x=0, y=0$ the result is two. turn $C=2$.

Rigorous Exercise

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}, A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$B^T B = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}, B^T c = -1$$

$$C = c - p_A - p_B \Rightarrow C = \frac{A^T c}{A^T A} - \frac{B^T c}{B^T B} B \Rightarrow \begin{bmatrix} 0 \\ 8 \end{bmatrix} - 0 + \frac{2}{3} \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$$

$$q_1 = \frac{1}{\sqrt{2}} A, q_2 = \frac{\sqrt{2}}{\sqrt{3}} B, q_3 = \frac{\sqrt{3}}{\sqrt{4}} C$$

Since $d = (1, 1, 1)$ is not a linear combination of $\{A, B, C\}$ or $\{q_1, q_2, q_3\}$, respectively, d must be perpendicular to these spaces.

As such, the vectors perpendicular to d will be in spaces $\{q_1, q_2\}$ and $\{A, B, C\}$ since they lie in the same space where they are not combinations of d and vice versa.

Not super-rigorous, but fine.

29. -All points are linearly dependent. Test: All points $(c_1, c_2, \dots, c_n)$ are linearly independent. Enh... The idea is here.

4.4

10. $Q = [q_1, q_2, q_3]$ $C = [c_1, c_2, c_3]$

11. $C^T Q = c_1 q_1 + c_2 q_2 + c_3 q_3 = 0$

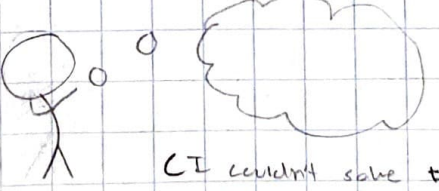
Since q_1, q_2, q_3 are orthonormal, $C_1^2 + C_2^2 + C_3^2 = 0 \Rightarrow C_1 = 0$

b. $Qx = 0$

By definition: $Q^T Q = I$ as $\|q_i\|^2 = 1$ for all $i \in \dim(Q)$

$\Rightarrow Q^T Q x = 0 \Rightarrow I x = 0 \Rightarrow x = 0$

THINKING SPACE



CI couldn't solve #10 for the life of me

35. [SEE CODE]

$$Q = \begin{bmatrix} -10 & -10 & -5 & 10 \\ 10 & 10 & -5 & 10 \\ 0 & 20 & -5 & 10 \\ 0 & 0 & 15 & 10 \end{bmatrix}$$

Differs by some factor C

5.1

$$10. \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \vec{x} = 0$$

Simply a vector with

$$\vec{x} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$
 since this

produces a dot product:

$a_{11} + \dots + a_{1n} = 0$ for all i gives the nullspace of A therefore contains

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \text{ meaning } N(A) \neq \{0\}$$

As such, A is not linearly independent $\Rightarrow$ not invertible.

If $\det(A-I) = 0$, $\det(A)$ is not necessarily 1. Changes of

invariants from other invariants

$$\text{determinants: } \det(A-I) \neq \det(A) - \det(I)$$

32. [SEE CODE]

rand: Random bit in the
range: $e+5, e+25, e+79, e+219$

randn: Random bit in the
range: $e+31, e+77, e+100, \dots$

I do not know what Inf is, but I assume it correlates to integer overflow as seen in [CODE]

$$36. A = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} P \\ 0 \end{bmatrix}$$

$Q_i: n \text{ cols} \rightarrow n$ - Square

$\text{rank}(A) = r$; $\text{rank}(P) = r$

$$r + (m-r) = m; \text{ beautiful!}$$

$$Q(A) = N(A)$$

$$\therefore Q_1 = Q(A), Q_2 = N(A)$$

$$\det \begin{bmatrix} 1 & a & a^2 \\ 0 & b & b^2 \\ 0 & c & c^2 \end{bmatrix}$$

$$\Rightarrow \det \begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{bmatrix}$$

$\Rightarrow$ I was very confused and I gave up after several tries.

$$(b-a) \det \begin{bmatrix} 1 & a & a^2 \\ 0 & c-a & c^2-a^2 \end{bmatrix}$$

$$= (b-a) \det \begin{bmatrix} 1 & a & a^2 \\ 0 & 1 & a+c \\ 0 & 0 & (c-a)(c+b) \end{bmatrix}$$

$$= (b-a)(c-a)(c-b)$$

33. [SEE CODE]

Therefore, parts are about:

$$1, 0.003, -0.006, 0.001, -0.000$$

Sols. used 4 sigfigs